SYS 599 : Thesis

Department
Systems Engineering
Credits 3.0
Prerequisites
Accepted to Systems Engineering M.S. degree program.

SYS 699 : Ph.D. Dissertation

Reserved for Ph.D. candidate research. Approval of the Ph.D. research advisor is required.
Department
Systems Engineering
Credits 0.0

Business School, The

BUS 500 : Business Law, Ethics and Social Responsibility

This course combines analysis of the structure, function and development of the law most important to the conduct of business with an examination of the ethical and social context in which managers make decisions. Emphasizing the social responsibility considerations of all business stakeholders, the course focuses on practical applications via extensive use of case studies. Students will gain a sound understanding of the basic areas of U.S. and international law including: intellectual property law; business formation and organization; international business law; securities regulation; cyber law and e-commerce; antitrust law; employment law and environmental law.
Department
Business School, The
Credits 3.0

BUS 522 : Global Business Experience

Business is increasingly global. To be successful one must understand the customs and traditions of the regions in which they are operating. This course provides students with insight into different countries and business environments and includes an international trip where students will spend a week to 10 days on the ground in the featured region meeting with business, government and/ or academic leaders; touring company sites; and learning about the region. Prior to the trip students will study business history, culture and current topics related to the featured region. Guest speakers will often be incorporated. Following the trip students will typically write reflective papers and deliver presentations.
Department
Business School, The
Credits 3.0

BUS 546 : Managing Technological Innovation

This course studies successful innovations and how firms must enhance their ability to develop and introduce new products and processes. The course will discuss a practical model of the dynamics of industrial innovation. Cases and examples will be discussed for products in which cost and product performance are commanding factors. The important interface among R&D/ manufacturing/marketing is discussed. International technology transfer and joint venture issues are also considered.
Department
Business School, The
Credits 3.0

BUS 547 : Energy Management

This course covers a broad spectrum of energy auditing methods, energy management planning and energy management topics important to future energy professionals, business managers and leaders. The course includes a project that applies energy management concepts to an actual energy audit. The audit project also includes the development of an energy management plan for a selected building making cost-effective recommendations to improve efficiency. Additional topics include: energy management strategies for business, governmental regulations, incentives and resources, European Union energy policies and programs including carbon credits and related markets. Energy efficiency practices as they relate to ISO 50001 Energy Management Systems, the U.S. Green Building Council (USGBC), and Leadership in Energy & Environmental Design (LEED), high performance buildings, data centers, renewable energy sources and smart grid are also studied. Special focus will be on energy management for financial and environmental sustainability benefits from the perspective of CEOs, CFOs, COOs and CSOs.

Department
Business School, The
Credits 3.0

BUS 590 : Strategic Management

This integrative and interdisciplinary course provides a broad overview of strategic management, with a focus on technology-driven organizations. Adopting a general management perspective, students will learn how to develop and execute a holistic corporate strategy that integrates key functional and business unit level strategies. Topics include data-driven strategy formulation, implementation, and evaluation. This course integrates the MBA core courses, and therefore should be taken after completing all core courses. It also serves as a prerequisite for the capstone project so it must be taken before the final capstone course (BUS 599).

Department
Business School, The
Credits 3.0
Prerequisites
FIN 500, MIS 584, MKT 500, OBC 506 and OIE 501 or equivalent content, or instructor consent

BUS 594 : Data-driven Business Strategy (Capstone)

This is the Capstone project course for the STEM MBA program. The course content will introduce students to business strategy and integrate it with data-driven decision making. The course will involve a major team-based analytics project with an organizational sponsor. The project will involve both technical analytical skills and strategic thinking for formulating and implementing effective business strategies.

Department
Business School, The
Credits 3.0

BUS 596 : Master of Science Capstone Project

This course is the capstone course for the STEM-based, specialty MS programs in the School of Business. This course serves as a practical integration of the STEM-based tools, techniques, and skills and the related business theories and practices that students learned in their MS program. The medium is a major team-based project in the form of an actual corporate STEM-based business need for which students will develop solutions. Students will produce a written report that documents and provides the financial, organizational, and technical rationale for the solutions. They will also formally present their results to the project sponsors. Students are expected to have completed (or are currently completing) all the courses requirements for their MS program prior to taking the capstone project. While the capstone requirements are the same for all STEM-based, specialty MS programs in the School of Business, the actual content of the project will differ by sponsor and by MS program. Students must take the appropriate section of BUS 596 for each MS program they complete.

Department
Business School, The
Credits 3.0
Prerequisites
MIS 502, MIS 584, OBC 503, OIE 552 or equivalent content, or instructor consent

BUS 597 : Independent Study

This course will allow a student(s) to study a certain business-related topic under the guidance of an affiliated WPI Business School faculty member. The student must produce an appropriate deliverable from this experience to satisfy the course requirement.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable

BUS 598 : Special Topics

This course will engage students at an advanced level in the exploration of special topics that reflect the expertise of the business school's faculty. It will serve as a flexible vehicle to provide a course offering of topics of current interest as well as to offer new topics before they are made into a permanent course. The course content and format vary to suit the interests and needs of the faculty and students.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable

BUS 599 : Capstone Project

This capstone course integrates management theory and practice, and incorporates a number of skills and tools acquired in the M.B.A. curriculum. The medium is a major team-based project in the form of a corporate venture or green field venture. In addition to a written report, the project is formally presented to a panel of outside experts including serial entrepreneurs and investors. (Students cannot get credit for BUS 599 and BUS 517)

Department
Business School, The
Credits 3.0
Prerequisites
BUS 590, FIN 500, MIS 584, MKT 500, OBC 506 and OIE 501 or equivalent content, or instructor consent

BUS 631 : Research Methods and the Research Process

This course introduces PhD students to business problems and the nature, scope, and purpose of research and research methodologies to study those problems. Topics include research approaches and designs, data types and their collection, measurement approaches, testing procedures, and interpreting and presenting findings. The ethics of various methods and data collection procedures are covered, as is the Institutional Review Board (IRB) process. Students will investigate applications of research methods to specific problems within their interest area, using qualitative and quantitative designs. They will also read example articles that use the research approaches covered.

Department
Business School, The
Credits 3.0
Prerequisites
Admission to the Business School PhD program or to another WPI PhD program.

BUS 632 : Qualitative Research Methods

Capturing and analyzing rich sets of qualitative (descriptive, non-numerical) data is typically required to investigate complex business cases, from discovering opportunities for innovation to identifying root causes of business problems. While quantitative business data is increasingly available, qualitative data is often the source of the deeper understanding needed for exploring, explaining, and developing theories for complex socio-technical business environments that dominate our digital economy. This course focuses on methods for collecting and analyzing qualitative research data with the purpose of enabling students to make the novel discoveries that characterize PhD research in business. Students will become familiar with methods for collecting (e.g., interviews and focus groups) and analyzing (e.g., coding techniques and case study analysis methods), as well as the software packages that aid in qualitative data organization and analyses. They will understand which collection, coding, and analysis techniques to use in which situations, how to interpret results, and how data collection and analysis methods affect research results. These skills are critical for understanding how business can contribute to solving the large, societal problems within and across disciplines and industries.

Department
Business School, The

Prerequisites
Admission to the Business School PhD program or to another WPI PhD program, BUS 631 or equivalent knowledge

Prerequisite Courses
BUS 631

BUS 633 : Quantitative Research Methods

Developing predictive behavioral models, which heavily rely on quantitative (numeric) data, are a major success factor in helping businesses develop competitive products and services. This course focuses on methods for collecting and analyzing quantitative research data with the purpose of enabling students to make the novel discoveries that characterize PhD research in business. Students will become familiar with one or more internationally utilized statistical software packages and with the array of statistical analysis techniques in them. They will understand which statistical analysis techniques to usein which situations, how to interpret the output from these packages, and how data collection and analysis methods affect research results. In our increasingly data intensive business environment, these skills are critical for understanding business data and using that understanding to design better processes and systems and to make better decisions within and across industries.

Department
Business School, The

Credits 3.0

Prerequisites
Admission to the Business School PhD program or to another WPI PhD program, BUS 631 or equivalent knowledge

BUS 651 : Seminar on Designing and Conducting Research Studies

This course is offered every semester for cohort students as they start their research studies. It bridges between students' methods courses and the start of their 30 dissertation credits. It is conducted in seminar format with a focus on students presenting the progress on their research studies and discussion among the class about appropriate research designs and analyses. This course can be taken multiple times.

Department
Business School, The

Credits 3.0

Prerequisites
Admission to the Business School PhD program or to another WPI PhD program, BUS 631 or equivalent knowledge, BUS 632 or BUS 633 or equivalent knowledge.

BUS 691 : Graduate Seminar

Seminars on current issues related to entrepreneurship, information technology and operations management are presented by authorities in their fields. All full-time Ph.D. students in Business Administration are required to register and attend.

Department
Business School, The

Credits 0.0

BUS 697 : Teaching Practicum

The objective of the teaching practicum is to provide students with practical teaching exposure within an actual classroom environment. This practicum requires students to closely collaborate with a faculty member to observe teaching methodologies and gain practical teaching exposure. The configuration of experiences and the scope of supervisory duties will be determined by the student and the advisor. Given WBS's mentorship-based doctoral program, this practicum is typically completed as an independent study under the supervision of the PhD student's advisor. During these practica, students are guided and assessed by their faculty supervisors. The nature of students' practicum experiences will differ, spanning factors such as the degree of focus on actual teaching sessions, the creation of course materials (e.g., syllabi, assignments, and discussions, etc.), grading assignments, as well as engaging in curriculum evaluation and planning. Enrollment in this course is not a teaching assistant assignment.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable
Prerequisites
Consent of research advisor

BUS 698 : Directed Research

For Ph.D. students wishing to gain research experience peripheral to their thesis topic.

Department
Business School, The
Credits Variable
Prerequisites
Consent of research advisor

BUS 699 : Dissertation Research

Intended for Ph.D. students admitted to candidacy wishing to obtain research credit toward their dissertations.

Department
Business School, The
Credits Variable
Prerequisites
Consent of research advisor

BUS 5900 : Internship

The internship is an elective-credit option designed to provide an opportunity to put into practice the principles that have been studied in previous courses. Internships will be tailored to the specific interests of the student. Each internship must be carried out in cooperation with a sponsoring organization, generally from off campus, and must be approved and advised by a WPI faculty member in the School of Business. Internships may be proposed by the student or by an off-campus sponsor. The internship must include proposal, design and documentation phases. Following the internship, the student will report on his or her internship activities in a mode outlined by the supervising faculty member. Students are limited to counting a maximum of 3 internship credits toward their degree requirements. Students must be making satisfactory academic progress as defined in the WPI graduate catalogue to be eligible to register for internship credit. International students who are working on a second U.S. masters degree and who have already used their master's-level Optional Practical Training (OPT) may petition the School of Business' Graduate Policy and Curriculum Committee to do additional Curricular Practical Training (CPT) beyond 3 credits on a non-credit basis. Part-time students cannot do an internship at their place of employment.

Department
Business School, The
Credits 0.0
-3

BUS 6900 : Internship for Ph.D. Students

Department
Business School, The
Credits 3.0

ETR 598 : Special Topics

This course will engage students at an advanced level in the exploration of special topics that reflect the expertise of the business school's ETR faculty. It will serve as a flexible vehicle to provide a course offering of topics of current interest as well as to offer new topics before they are made into a permanent course. The course content and format vary to suit the interests and needs of the faculty and students.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable

FIN 598 : Special Topics

This course will engage students at an advanced level in the exploration of special topics that reflect the expertise of the business school's FIN faculty. It will serve as a flexible vehicle to provide a course offering of topics of current interest as well as to offer new topics before they are made into a permanent course. The course content and format vary to suit the interests and needs of the faculty and students.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable

MIS 598 : Special Topics

This course will engage students at an advanced level in the exploration of special topics that reflect the expertise of the business school's MIS faculty. It will serve as a flexible vehicle to provide a course offering of topics of current interest as well as to offer new topics before they are made into a permanent course. The course content and format vary to suit the interests and needs of the faculty and students.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable

MKT 598 : Special Topics

This course will engage students at an advanced level in the exploration of special topics that reflect the expertise of the business school's MKT faculty. It will serve as a flexible vehicle to provide a course offering of topics of current interest as well as to offer new topics before they are made into a permanent course. The course content and format vary to suit the interests and needs of the faculty and students.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable

OBC 598 : Special Topics

This course will engage students at an advanced level in the exploration of special topics that reflect the expertise of the business school's OBC faculty. It will serve as a flexible vehicle to provide a course offering of topics of current interest as well as to offer new topics before they are made into a permanent course. The course content and format vary to suit the interests and needs of the faculty and students.

Department
Business School, The
Credits 1.0
-3 **Credits** Variable

Entrepreneurship

ETR 500 : Entrepreneurship and Innovation

Entrepreneurship involves many activities, including identifying and exploiting opportunities, creating and launching new ventures, introducing new products and new services to new markets. It is based on implementing innovations within existing organizations and creating new opportunities. This course is intended to introduce students to entrepreneurial thinking and methods of executing their ideas. Topics include recognizing and evaluating opportunities, forming new venture teams, preparing business and technology commercialization plans, obtaining resources, identifying execution action scenarios, and developing exit strategies.

Department
Entrepreneurship
Credits 3.0

ETR 593 : Technology Commercialization

In the modern world of global competition the ability to utilize technological innovation is increasingly important. This course will examine the sources of new technology, the tools to evaluate new technologies, the process of intellectual property transfer, and the eventual positioning of the resultant products and services in the commercial market. Its purpose is to improve the probability of success of this discipline in both existing organizational models and early stage ventures. Specific cases studies of successful technology commercialization processes will be used to supplement the course materials.

Department
Entrepreneurship
Credits 3.0

ETR 596 : Selling and Sales

Selling is a major part of our business and professional lives. This is especially important for those who are launching new ventures. Business propositions need to be presented to (and need to be sold to) potential investors, employees, colleagues, and certainly potential employers. Later there is a need to sell products or services to customers. Common to all is a sales process and organization model that can be developed that is focused on meeting customer and other stakeholder needs through effective selling disciplines.

Department
Entrepreneurship
Credits 3.0

Finance

FIN 500 : Financial Management

This course develops students' financial expertise. The course focuses on financial management and corporate finance. Students learn accounting and financial concepts, principles, and methods for preparing, analyzing, and evaluating financial information, for the purpose of managing financial resources of a business enterprise and making investment decisions. Students are also introduced to the principles and methods of valuation. Students practice with the financial reporting system which enables data analysts to build queries for financial analyses and to forecast possible financial scenarios. Finally, this course focuses on financial strategy and planning to enable internal managerial decisions. Students will learn and apply budgeting techniques and manage working capital.

Department
Finance
Credits 3.0

FIN 503 : Financial Decision-Making for Value Creation

This course develops and enhances the student's ability to implement and clearly communicate a firm's financial decisions related to value creation. The course covers capital structure optimization, cost of capital; capital allocation and investment strategies, enterprise risk, project and firm valuation, and international financial management. The course adopts a decision-maker's perspective by emphasizing the relationships among a firm's strategic objectives, financial accounting and financial statement data, economic events, responses by market participants and other impacted constituencies, and corporate finance theory. The course also builds on these practical finance skills by incorporating team-based assignments, real-world simulations, and a variety of financial modeling tools. (Students cannot get credit for FIN 503 and FIN 500)

Department

Finance

Credits 3.0

FIN 510/MIS 510 : Business Application of Blockchain Technology

Business Applications of Blockchain Technology provides students with a foundation of blockchain technology examining its applications across various disciplines such as engineering, law, and economics. Throughout the course, students will delve into diverse blockchain and distributed ledger technologies, popular cryptocurrencies such as Bitcoin and Ethereum, and the implementation and security concerns surrounding them. Students will learn how these systems work, explore the regulatory challenges associated with blockchain technology, and assess its impact on financial services and other industries. The course will also provide students with the component and structures of blockchain business networks, such as smart contracts, consensus, certificate authorities, security protocols, role of nodes, transaction processes, participants, and underlying frameworks. Furthermore, the course will examine the Bitcoin (BTC) ecosystem, Ripple (XRP), Ethereum (ETH), tokens (e.g., NFT's), Initial Coin Offerings (ICOs), and Central Bank Digital Currencies (CBDC). Students will also gain hands-on experience, such as creating basic Blockchain contracts, enabling them to discuss the practical capabilities and limitations of blockchain technology.

Department

Management Information Systems

Finance

Credits 3.0

FIN 521 : Financial Management in a Global Environment

This course builds from Financial Information and Management, and extends closed-economy financial management to the international market environment. Drawing from theories based on culture, corporate finance, and investor protection laws, this course examines differences in corporate governance, financial information, and financial markets in global settings. The first focus is on accountability of financial resources, the implications of globalization on firms' financial reporting and decision-making. The second focus is on international markets and institutions, how the access and exposure to different market environments can affect the firm's financial and investment decisions. Major topics include the relationship between foreign exchange and other financial variables; measurement and management of the exchange risk exposure of the firm; international investment decisions by firms and investors; and financing the global operations of firms.

Department

Finance

Credits 3.0

FIN 522 : Financial Institutions, Markets & Technology

This course will examine financial institutions and the relationship between U.S. capital markets and global markets. The class is intended to help students understand the impact of financial intermediaries on the global economy, businesses, and consumers. The course will investigate the organization, structure, and performance of money and capital markets and institutions. The class will examine the major financial management issues confronting financial service firms (depository institutions, insurance companies, investment banks, mutual funds, hedge funds, and pension funds), and it will address the legal, regulatory, financial reform, and risk management issues facing these financial institutions and markets. Finally, the course will address the rapid evolution of the financial sector as a result of technology. We will consider how financial technology ("FinTech") is being developed by startup technology firms and existing financial institutions may disrupt the financial sector through innovation in digital and electronic currencies, online finance and investment platforms, big data, and digital payment systems among other topics.

Department

Finance

Credits 3.0

FIN 530 : Cryptocurrencies and Financial Markets

This course covers digital currencies and related topics in the FinTech area. The course begins with studying the nature of money, legacy payment, and banking systems. The course then examines the emergence of stateless, cloud-based digital currency systems since 2009. Students will also gain insight into the functioning of decentralized assets in today's financial markets and the role of fintech assets such as cryptos in financial intermediation. Students will learn about central bank digital currencies and how they will help to improve banking by reducing the under- banked and un-banked population. The course will also emphasize the inherent risks and ethical dilemmas and regulatory compliance challenges (for example, Anti-Money Laundering and Know Your Customer regulations) of this emerging asset class.

Department

Finance

Credits 3.0

FIN 540 : Financial Analytics

The course introduces advanced methodological tools required for conducting finance and investment analysis research. The course aims to equip students with a working knowledge of important econometric techniques used in financial economics, such as event study, advanced time series analysis, and survival analysis. Substantial emphasis will be placed on developing programming skills in computer programs. The course emphasizes understanding and learning how to apply practitioners' econometric tools in these areas. Students will also cover the basic theory of statistical inference with linear models, general linear models, Heteroskedasticity models, time series models, analysis of variance, discriminate analysis, factor analysis, and non-parametric tests

Department

Finance

Credits 3.0

FIN 698 : Directed Research

FIN 698 is administered by the Director of the financial technology program, with the support of faculty members who possess specialized expertise in their respective research fields. This Directed Research course is specifically designed for Ph.D. students aiming to expand their research skills and specialization along the lines of their dissertation research. It offers an exceptional opportunity for these students to conduct comprehensive, faculty-supervised research in various finance-related areas, thereby enhancing their grasp of advanced financial theories and their practical applications.

Department

Finance

Credits Variable

Management Information Systems

FIN 510/MIS 510 : Business Application of Blockchain Technology

Business Applications of Blockchain Technology provides students with a foundation of blockchain technology examining its applications across various disciplines such as engineering, law, and economics. Throughout the course, students will delve into diverse blockchain and distributed ledger technologies, popular cryptocurrencies such as Bitcoin and Ethereum, and the implementation and security concerns surrounding them. Students will learn how these systems work, explore the regulatory challenges associated with blockchain technology, and assess its impact on financial services and other industries. The course will also provide students with the component and structures of blockchain business networks, such as smart contracts, consensus, certificate authorities, security protocols, role of nodes, transaction processes, participants, and underlying frameworks. Furthermore, the course will examine the Bitcoin (BTC) ecosystem, Ripple (XRP), Ethereum (ETH), tokens (e.g., NFT's), Initial Coin Offerings (ICOs), and Central Bank Digital Currencies (CBDC). Students will also gain hands-on experience, such as creating basic Blockchain contracts, enabling them to discuss the practical capabilities and limitations of blockchain technology.

Department

Management Information Systems

Finance

Credits 3.0

MIS 500 : Innovating with Information Systems

This course focuses on information technology and innovation. Topics covered are information technology and organizations, information technology and individuals (privacy, ethics, job security, job changes), information technology and information security, information technology within the organization (technology introduction, and implementation, and data analytics for competitive analysis), business process engineering and information technology between organizations (electronic data interchange and electronic commerce). This course provides the knowledge and skills to utilize existing and emerging information technology innovatively to create business opportunities.

Department
Management Information Systems
Credits 3.0

MIS 502 : Data Management for Analytics

This course develops the skills business students need for handling data. It focuses on student skills in (1) cleaning and preparing data for analysis, (2) writing SQL queries to access and manipulate data, and (3) ethical uses of data and data privacy issues. It also covers the types of data typically found in organizations, e.g., employee, customer, product, marketing, operations, and financial data.

Department
Management Information Systems
Credits 3.0

MIS 520 : Artificial Intelligence and its Ethical Applications in Business

This course provides students with a comprehensive introduction to the recent developments in AI, with a focus on their current and potential practical applications in business. The course covers fundamental AI concepts such as search methods, logic, knowledge representation, and natural language processing in sufficient detail to understand how these concepts and associated tools can be applied to solve practical business problems in ways that create fair and ethical solutions. Students will learn about bias in data, algorithms, and human decision making, and methods of avoiding the transfer of those biases to AI-based solutions. They will gain practical experience though a project about the opportunities of AI-based solutions to assist with making decisions and predictions and transforming business and work, as well as the challenges of doing so in a fair and ethical manner that benefits individuals, the environment, and society at large.

Department
Management Information Systems
Credits 3.0

MIS 571 : Database Applications Design and Development

This course introduces students to the theory and practice of computer-based data management, including the delivery of high quality data in information processing and analysis. The course focuses on the design of database systems to meet an organization's needs for data analytics. The course also covers data security, data integrity, data quality, as well as backup and recovery procedures. Students will be exposed to commercially available database management systems, such as Microsoft Access and Oracle. As a project during the course, students will design and implement a small database application that meets the data needs of some real-world business opportunity. The project report will include recommendations for ensuring data security, data integrity, and data quality.

Department
Management Information Systems
Credits 3.0

MIS 573 : System Design and Development

This course introduces students to the concepts and principles of systems analysis and design. It covers all aspects of the systems development life cycle from project identification through project planning and management, requirements identification and specification, process and data modeling, system architecture and security, interface design, and implementation and change management. Object-oriented analysis techniques are introduced. Students will learn to use an upper level CASE (computer-aided software engineering) tool, which will be employed in completing a real-world systems analysis and design project.

Department
Management Information Systems
Credits 3.0

MIS 576 : Project Management

This course presents the specific concepts, techniques and tools for managing projects effectively. The role of the project manager as team leader is examined, together with important techniques for controlling cost, schedules and performance parameters. Lectures, case studies and projects are combined to develop skills needed by project managers in todays environment.

Department

Management Information Systems

Credits 3.0

MIS 581 : Policy and Strategy for Information Technology, Artificial Intelligence, and Analytics

This course focuses on the core IT capabilities that IT executives and managers must consider when creating strategy and policy for the deployment and use of information technologies, AI tools and techniques, and data within an organization. Those IT capabilities include both operational uses of such technologies, tools, and data, as well as strategic decision making for deploying advanced AI and Analytics capabilities. The expanding role of IT executives to encompass AI and Analytics capabilities as well as the fast-paced changes in technology require IT executives and managers to quickly understand, adapt, and apply technology that can provide value to organizations. Successful organizations are those that can react quickly by introducing innovative technologies and respond to market demands using data driven solutions produced by the ethical use of AI and Analytics models and tools. Students will learn how IT managers engage data and AI tools ethically to develop and enhance their departments' strategies.

Department

Management Information Systems

Credits 3.0

MIS 582 : Information Security Management

This course introduces students to the fundamentals of Information Security Management. It is designed to develop in students an understanding of and appreciation for the importance of information security to all enterprises, and to enable current and future managers to understand the important role that they must play in securing the enterprise. This course is appropriate for any student interested in gaining a managerial-level understanding of information security. A combination of readings, lectures, case studies, guest speakers, and discussion of real world events will be used to bridge the gap between theory and practice. The course will primarily explore the Common Body of Knowledge (CBK) of information security, along with other related topics. It will also explore the interaction between People, Process and Technology as the cornerstone of any effective information security program. Upon completion of this course, the student will have an in-depth understanding of the essential components of a comprehensive information security program, as well as an understanding of the technology at work behind the scenes.

Department

Management Information Systems

Credits 3.0

MIS 583 : User Experience Applications

The UX Applications course provides an introduction to using UX methods to study user experience. The course teaches students how to use the newest research tools, including eye tracking and emotion detection, to study user experiences of technological products and services. Students will learn how businesses can benefit from these techniques. Both theoretical concepts and practical skills will be addressed within the scope of the class through hands-on projects, class exercises, and assignments.

Department

Management Information Systems

Credits 3.0

MIS 584 : Business Intelligence

This course provides students with the knowledge and skills to design, develop, and use business dashboards for monitoring organizational performance and making data-driven decisions. On the technical side, students will learn and apply business intelligence software to organize, represent, and analyze data about customers, products, sales, marketing, operations, and financials. They will learn to create strategic, operational, and analytical dashboards displaying key performance indicators (KPIs) for managerial decision-making. On the business side, students will learn the connections between business strategy and plans, the KPIs that measure performance compared to those plans, and how to use dashboards to manage organizational performance. Students will also learn the technical and managerial challenges of creating and deploying these business intelligence best practices so that organizations gain value from their data. The course includes business cases and hands-on analyses of business data. It is designed for any student interested in learning about data-driven business performance management, including students whose primary focus is Business Management, Data Science, IT, Marketing, or Operations.

Department
Management Information Systems
Credits 3.0

MIS 585 : User Experience Design

Creating value through designing positive user experiences is increasingly important for staying competitive in the marketplace. This course covers best practices for designing innovative user-centered product and service experiences that resonate with users. Through a series of hands-on projects, students will develop practical skills in showcasing the inherent value of their user-centered designs to stakeholders. Emphasis will be placed on constructing portfolios that effectively demonstrate the value of their designs and translate their design outcomes into tangible business results.

Department
Management Information Systems
Credits 3.0

MIS 586 : User Experience Methods

In today's digital economy, understanding how people use and experience technology is crucial to designing successful technological products and services. This course covers the methodologies and tools for conducting research in the User Experience (UX) field. The course covers both qualitative and quantitative methods for conducting UX research in academia and industry, including surveys, persona development, customer journey maps, and other industry-standard tools for studying user experience. Both cutting-edge theoretical concepts and proven practical skills will be addressed within the scope of the class through hands-on projects, class exercises, and assignments.

Department
Management Information Systems
Credits 3.0

MIS 587 : Business Applications in Machine Learning

This course explores how Machine Learning (ML) and Artificial Intelligence (AI) is applied to solve business problems, to satisfy specific business needs, or to discover new opportunities for businesses. Applications of ML and AI are constantly evolving across many industries. This course utilizes existing AutoML solutions to address issues identified in business case studies (e.g. predicting hospital readmissions, loans likely to default, customer churn). The course covers the machine learning project life cycle starting with defining ML project objectives, acquiring and exploring data, modeling using AutoML tools, interpretation of models and communication of outcomes, and implementation and deployment of predictive models in organizations.

Department
Management Information Systems
Credits 3.0

Marketing

MKT 500 : Marketing Strategy

This course enables students to draw insights from data to formulate effective marketing strategies that benefit the organization and its stakeholders. Students will learn to (1) identify and understand consumers' value needs (marketing research and consumer behavior), (2) create an attractive value proposition (product and pricing strategies and tactics), and (3) communicate and deliver this value proposition (promotion and distribution strategies and tactics). Upon successful completion of this course, students will be able to develop and execute an effective data-driven marketing plan to achieve an organization's financial and marketing goals. Experiential learning techniques will be used to impart this knowledge and develop these skills.

Department
Marketing
Credits 3.0

MKT 561 : Consumer Behavior and Analytics

We are living in a data-driven world. Everything we do from getting our news in the morning, to buying goods, and searching for information leaves trails of data across the Internet. Consumers have changed and companies need to find new ways to engage with consumers in order to stay profitable and relevant. As a working professional, you will be tasked to use data to make business decisions and develop strategy that create value for consumers and your organization. This course will introduce traditional theories of consumer behavior and then take you on a beginning journey through the dynamic practices of how to use consumer data and analytics in the digital age. Topics covered include consumer behavior theory, an examination of attitude formation and value creation, the challenges of consumer protection, market research, and the influence of technology on consumer decision making.

Department
Marketing
Credits 3.0

MKT 562 : Marketing Research

This course is designed to equip students with research methods and tools that are used for marketing decision making. Students will learn to conduct, use, apply, interpret, and present marketing research in order to become effective decision makers. The topics covered in this course include problem formulation, research design, data collection methods, data analysis, and finally presentation of a research plan. This course will be an activity-based course involving design, implementation, and presentation of a marketing research plan. Basic knowledge of marketing and statistical concepts is assumed.

Department
Marketing
Credits 3.0

MKT 565 : Digital and Social Media Marketing

The rapid evolution of technology has led to increasingly well-informed buyers who are connected, communicative, and more in control than ever. This course discusses digital technologies and their impact on the marketing mix, branding, communication strategies, and distribution channels. Emphasis is placed on contemporary topics that face today's marketing managers –with a focus on how social media can be employed to build brands, conduct business, support causes, rally the masses, and create and maintain customer relationships. Students who have previously taken MKT 565 (Digital Marketing) or MKT 598 (Special Topics: Social Media Marketing) cannot earn credit for taking MKT 565 (Digital and Social Media Marketing).

Department
Marketing
Credits 3.0

MKT 568 : Marketing Analytics

Data is at the heart of this new era of marketing. The goal of this course is to provide the skills needed to make intelligent use of marketing data about customers, competitors, and the industry. The focus will be on the application of analytics techniques to enhance marketing making in organizations. The course blends the art and science of marketing and prepares students to generate marketing insights from data in areas such as segmentation, targeting, positioning, product choice, customer satisfaction, and customer lifetime value analysis. This will be a hands-on course, in which students apply the concepts and techniques studied in class to actual business situations.

Department
Marketing
Credits 3.0

MKT 569 : Product Management

A successful product management process involves vision, strategy, and product development and integrating these with an effective go-to-market strategy. In this project-based course, students will develop knowledge of product management concepts and frameworks, learn to work with product management tools, and build the skills necessary to become effective product managers.

Department
Marketing
Credits 3.0

Operations and Industrial Engineering

OIE 501 : Operations Management

This course focuses on the data-driven decision-making that matches supply to demand in an organization and its supply chain, emphasizing the strategic impact of operations on competitiveness and sustainability. Emergent technologies are explored as opportunities for innovation. Descriptive, predictive, and prescriptive analytical techniques are introduced to structure and evaluate key operational decisions. Skills required to model a system's operations, to address uncertainty and mitigate risk, to effectively evaluate resource needs, to integrate components into a coordinated system, and to efficiently develop and manage capacity and inventory are honed during the course.

Department
Operations and Industrial Engineering
Credits 3.0

OIE 542 : Risk Management and Decision Analysis

Risk management deals with decision making under uncertainty. It is interdisciplinary, drawing upon management science and managerial decision-making, along with material from negotiation and cognitive psychology. Classic methods from decision analysis are first covered and then applied, from the perspective of business process improvement, to a broad set of applications in operations risk management and design including: quality assurance, supply chains, information security, fire protection engineering, environmental management, projects and new products. A course project is required (and chosen by the student according to his/her interest) to develop skills in integrating subjective and objective information in modeling and evaluating risk. (Students cannot get credit for both OIE 542 and OIE 541

Department
Operations and Industrial Engineering
Credits 3.0

OIE 544 : Supply Chain Analysis and Design

This course studies the decisions, strategies and analytical methods in designing, analyzing, evaluating, and managing supply chains. Concepts, techniques, and frameworks for better supply chain performance are discussed, and how digital technologies enable companies to be more efficient and flexible in their internal and external operations are explored. The major content of the course is divided into three modules: supply chain integration, supply chain decisions, and supply chain management and control tools. Students will learn how to apply some of the techniques in Operations Research such as linear programming, dynamic programming, and decision tree to aid decision-making. A variety of instructional tools including lectures, case discussions, guest speakers, games, videos, and group projects and presentations are employed.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 548 : Performance Analytics

Productivity management and performance analysis techniques and applications are covered from engineering and management perspectives. Topics include benchmarking, production functions, and the concept of relative efficiency and its measurement by data envelopment analysis. Application examples include efficiency evaluations of bank branches, sales outlets, hospitals, schools and others.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 549 : Sustainable Supply Chain and Operations Management

The environmental implications and responsibilities of organizations begin at an organization's boundaries with management of their operations, but also extend to incorporate interorganizational relationships and networks, the supply chain. We will investigate the practice and theory of sustainable supply chains and operations management in organizations throughout the world. This course is intended to provide students with understanding the intra- and interorganizational implications of environmental sustainability practices and policies. The role of organizational supply chain management functions, activities, tools and methods and their relationship to the natural environment will be introduced and discussed. The goals are for students to grasp the scope of general supply chain/operations management and environmental sustainability as they relate to the firm; to be able to relate to the manners in which management may respond and collaborate internally and with suppliers, customers, and various other stakeholders influencing and influenced by operational and supply chain activities from practical and theoretical case studies; able to evaluate various factors and understand tradeoffs in management decisions as they pertain to environmental supply chain management.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 552 : Modeling and Optimizing Processes

This course is designed to provide students with a variety of quantitative tools and techniques useful in modeling, evaluating and optimizing operation processes. Students are oriented toward the creation and use of spreadsheet models to support decision-making in industry and business.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 553 : Global Purchasing and Logistics

This course aims to develop an in-depth understanding of the decisions and challenges related to the design and implementation of a firm's purchasing strategy within a context of an integrated, global supply chain. Topics centering on operational purchasing, strategic sourcing, and strategic cost management will be covered. The global logistics systems that support the purchasing process will be analyzed, and the commonly used techniques for designing and evaluating an effective logistics network will be studied.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 554 : Global Operations Strategy

This course focuses on operations strategy from a global perspective. Topics such as strategy of logistics and decisions to outsource are examined. As an example, the strategic issues concerned with firms that are doing R&D in the United States, circuit board assembly in Ireland and final assembly in Singapore. Cases, textbooks and recent articles relating to the topic are all used. Term paper based on actual cases is required.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 557 : Service Operations Management

Successful management of service organizations often differs from that of manufacturing organizations. Service business efficiency is sometimes difficult to evaluate because it is often hard to determine the efficient amount of resources required to produce service outputs. This course introduces students to the available techniques used to evaluate operating efficiency and effectiveness in the service sector. The course covers key service business principles. Students gain an understanding of how to successfully manage service operations through a series of case studies on various service industries and covering applications in yield management, inventory control, waiting time management, project management, site selection, performance evaluation and scoring systems. The course assumes some familiarity with basic probability and statistics through regression.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 558 : Designing and Managing Lean Six Sigma Processes

In this course, Lean Six Sigma is presented as an organizational improvement system and a set of process analysis and statistical tools that have helped the world's leading organizations achieve operational excellence, saving millions of dollars and improving customer satisfaction. This course is organized in three parts: part one covers the essentials of Lean Six Sigma, including fundamental concepts and problem-solving methods; part two of the course covers Lean Six Sigma tools, including topics such as value-stream mapping, process capability, and experimental design; part three describes the major activities in a Lean Six Sigma roadmap, from identifying core processes to executing improvement projects to sustaining Lean Six Sigma gains.

Department

Operations and Industrial Engineering

Credits 3.0

OIE 559 : Advanced Prescriptive Analytics: From Data to Impact

This course provides an in-depth focus on prescriptive analytics, which involves the use of data, assumptions, and mathematical modeling of real-world decision problems to ascertain and recommend optimal courses of action. Starting from conceptualization of the problem, to using theory for translational modeling and techniques, to computational solving, and finally interpretation – likely in an iterative manner – students will gain knowledge of tools and practical skills in transforming real-world decision problems into actionable insights. Advanced topics in the prescriptive analytics domain will be covered, such as the use of integer variables to represent important logical constructs, using nonlinear functions to represent real-world decision aspects, the incorporation of stochasticity and uncertainty, and corresponding solution methods. Real-world problems will be selected from a variety of contexts that may include capacity management, data science, finance, healthcare, humanitarian operations, inventory management, production planning, routing, staffing, and supply chain. Students will complete an individual project that includes a report in the style of a technical report or research paper, as well as an oral presentation. Students may not receive credit for both OIE 4430 and OIE 559

Department

Operations and Industrial Engineering

Credits 3.0

Prerequisites

OIE 552, equivalent knowledge about optimization and linear programming, or consent of the instructor.

OIE 597 : Operations and Supply Chain Consulting Project

This capstone course serves as a practical integration of the operations and supply chain theories, practices, tools and techniques that students learned in their MS program. The medium is a major team-based project, sponsored by an external organization. The course goals are: (1) to enrich students' experiential learning and support the acquisition of the skills and capabilities to tackle real-world problems; and (2) to enhance students' teamwork, interpersonal and consulting skills. Students will produce a written report documenting their solutions, and providing the financial, organizational, and technical rationale for their approach. They will also formally present their results to the project sponsors. Students are expected to have completed (or be currently completing) all the course requirements for their MS in Operations and Supply Chain Analytics prior to taking the capstone project.

Department
Operations and Industrial Engineering
Credits 3.0
Prerequisites
OIE 501, OIE 544, OIE 552, OBC 505 or equivalent content, or instructor consent

OIE 598 : Special Topics

Department
Operations and Industrial Engineering
Credits 3.0

OIE 599 : Supply Chain Research

This research study is at the master's level. The course provides a research experience for students interested in studying a pressing supply chain management problem or challenge. Students must satisfactorily complete a written paper and are encouraged to publically present the results.

Department
Operations and Industrial Engineering
Credits 3.0

Organizational Behavior and Change

OBC 505 : Teaming and Organizing for Innovation

How do we navigate complex human systems in organizations? How do we foster innovation within organizations? In this course, we explore the paradoxes, opportunities, and hidden systemic challenges that arise on teams and projects, and in working across networks and within innovative organizations. Students will learn to more deftly manage the inherent challenges and opportunities of cross-cultural and multi-disciplinary teams; work through or avoid dysfunctional team and organizational conflicts; wrestle with ambiguity and uncertainty; negotiate change by learning to work with networks of power and influence; and analyze the individual, group, organizational and contextual dynamics that enable and constrain productive and innovative work in organizations. (Students cannot get credit for OBC 505 and OBC 500)

Department
Organizational Behavior and Change
Credits 3.0

OBC 506 : Leadership

How do we mobilize our own and others' energy toward developing sustainable outcomes and meaningful change—when the path ahead is unclear, when our business environment is rapidly changing, when we do not have full authority over those involved? This course embraces a human-centered design approach to leading others with integrity, empathy, and curiosity—with a specific focus on the unique challenges and opportunities of working within project-based networks and Industry 4.0/STEM contexts. Students will build their capacity to navigate complex human and technical systems as they work in teams to develop and pilot a solution to a real-life organizational or social problem.

Department
Organizational Behavior and Change
Credits 3.0

OBC 533 : Negotiations

This course focuses on improving the student1 s understanding of the negotiation process and effectiveness as a negotiator. Emphasizes issues related to negotiating within and on behalf of organizations, the role of third parties, the sources of power within negotiation, and the impact of gender, culture and other differences. Conducted in workshop format, combining theory and practice.

Department

Organizational Behavior and Change

Credits 3.0

OBC 535 : Managing Creativity in Knowledge Intensive Organizations

This course considers creativity in its broadest sense from designing new products and processes to creating our own role and identity as managers and leaders in knowledge-intensive organizations. In this course we will look actively at our own creative process and how we might more fully realize our creative potential. At the same time we will build a conceptual understanding of creating, creativity, and knowledge based in the philosophic, academic, and practitioner literatures. We will critically apply this conceptual understanding to organizational examples of managing creativity in support of practical action.

Department

Organizational Behavior and Change

Credits 3.0

OBC 536 : Organizational Design

A key role for organizational leaders is to design their organization to achieve their desired results. This course applies design thinking and methods to the practical problems of designing various sized organizations for optimal results in a complex environment. This is based on a foundation of organizational theory, design methodology, and organizational strategy.

Department

Organizational Behavior and Change

Credits 3.0

OBC 537 : Leading Change

This course focuses on the role of leadership in the design and implementation of organizational change. Topics include visioning, communication, social influence, power, resiliency, and resistance to change. Teaching methods include classroom discussion of readings and cases, simulations, and experiential exercises.

Department

Organizational Behavior and Change

Credits 3.0

OBC 538 : Developing Managerial Talent

Assessing and developing managerial talent in yourself and others is a key to professional success and can be a source of organizational competitive advantage. This course addresses the Globalizing World and You, and provides students access to the frameworks, tools, and practice necessary to engage in thoughtful self-assessment, constructive feedback acquisition and interpretation, and strategic development planning for themselves as well as for others on their teams and in their organizations. The goals of this course are: a) to help students assess their own managerial abilities, b) to develop plans for securing new knowledge, skills and abilities that will help them in their careers, c) to set goals and agendas for their own development and d) to consider ways to translate this development process to others.

Department

Organizational Behavior and Change

Credits 3.0